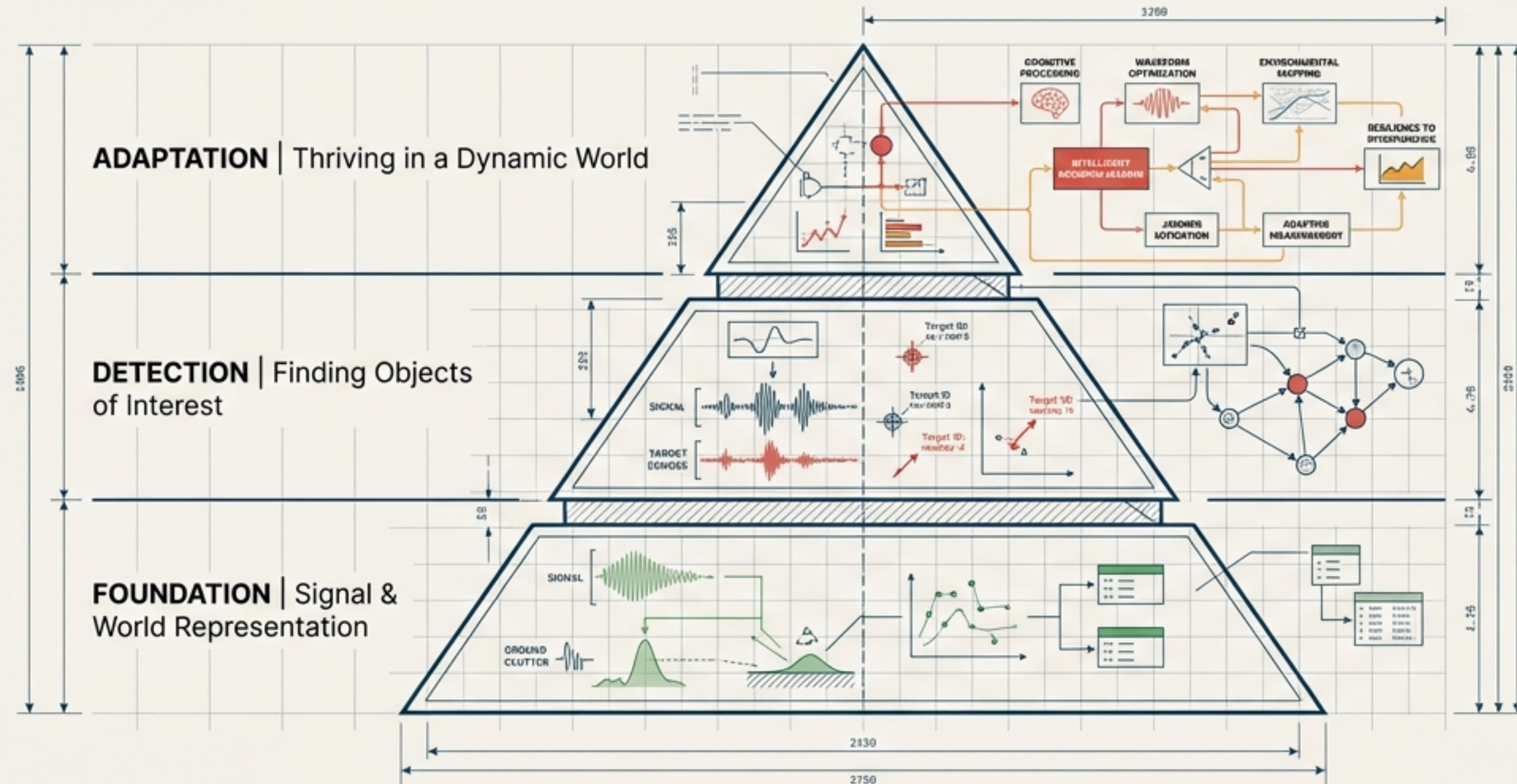


BUILDING THE MODERN RADAR SYSTEM

A Pyramid of Signal Processing Capability



This presentation deconstructs the algorithmic stack of a modern radar, showing how each layer builds upon the last to create a progressively more intelligent and resilient system.

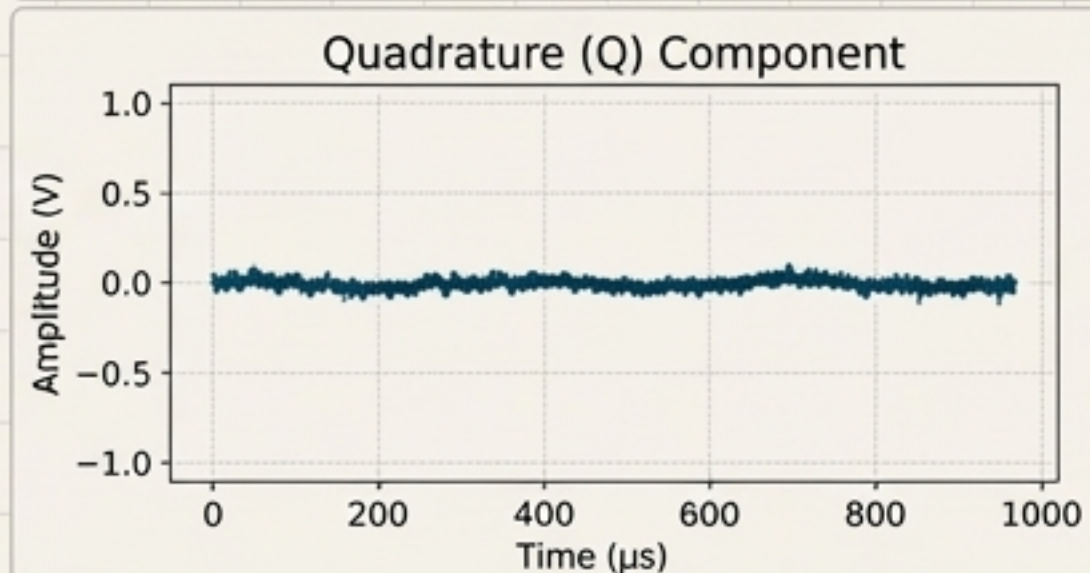
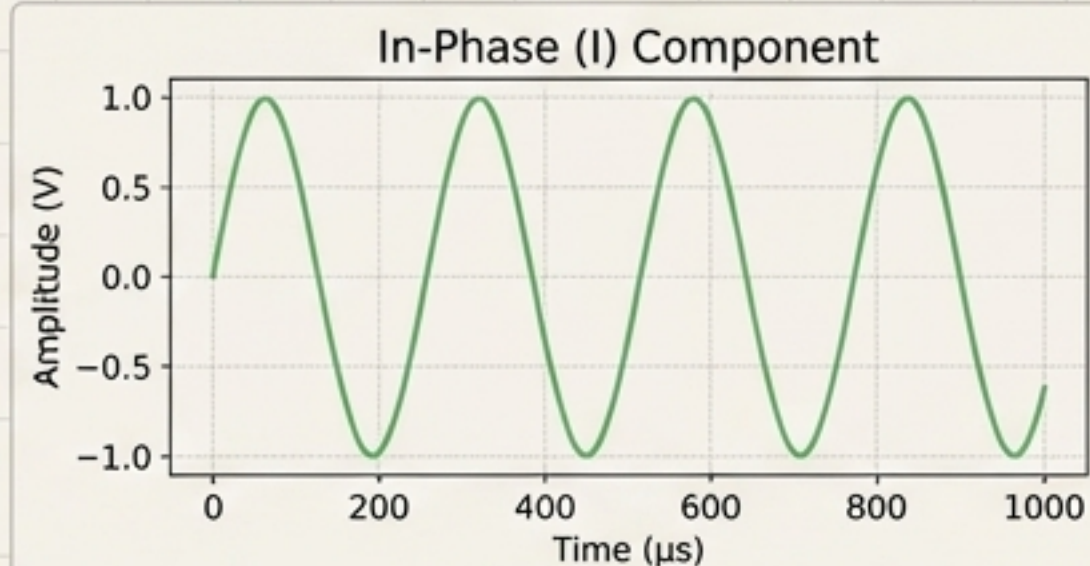
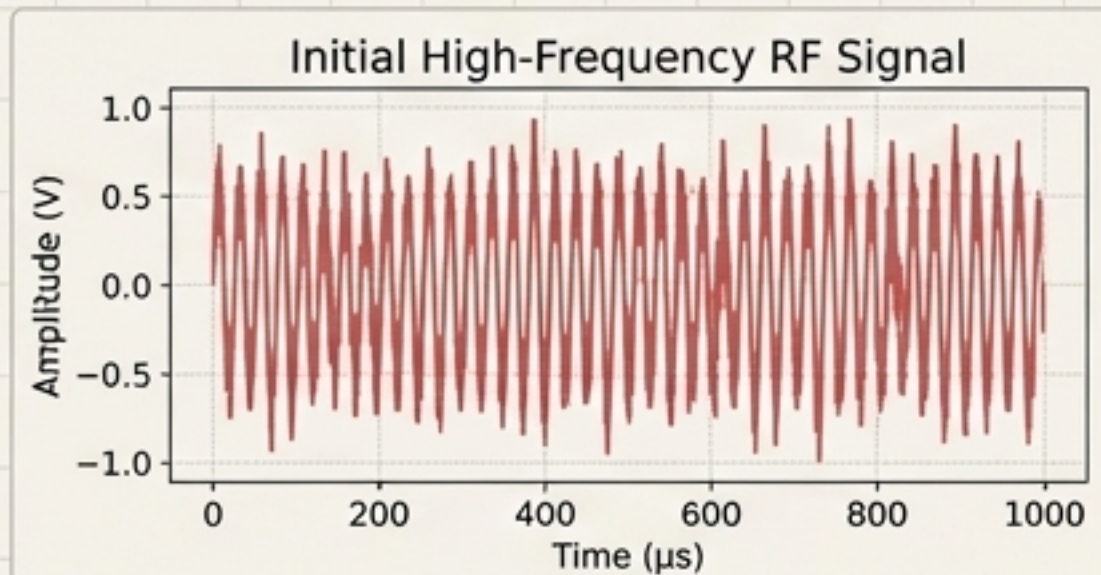
The Foundation: Taming RF Chaos with I/Q Representation

The Challenge

Direct processing of RF signals in the GHz range is computationally prohibitive. A raw signal, $s_{\text{RF}}(t) = A(t) \cos(2\pi f_c t + \phi(t))$, requires massive sampling rates and expensive hardware.

The Solution

We extract the complex baseband signal: $s_{\text{baseband}}(t) = I(t) + j \cdot Q(t)$. This shifts the signal to zero frequency while preserving all amplitude and phase information. The In-Phase (I) and Quadrature (Q) components modulate orthogonal cosine and sine carriers, allowing them to be separated perfectly.



By decomposing the raw RF signal into I and Q, we convert an intractable high-frequency problem into a manageable baseband computation.

The Peril of Imperfection: Diagnosing and Correcting I/Q Imbalance

The Challenge

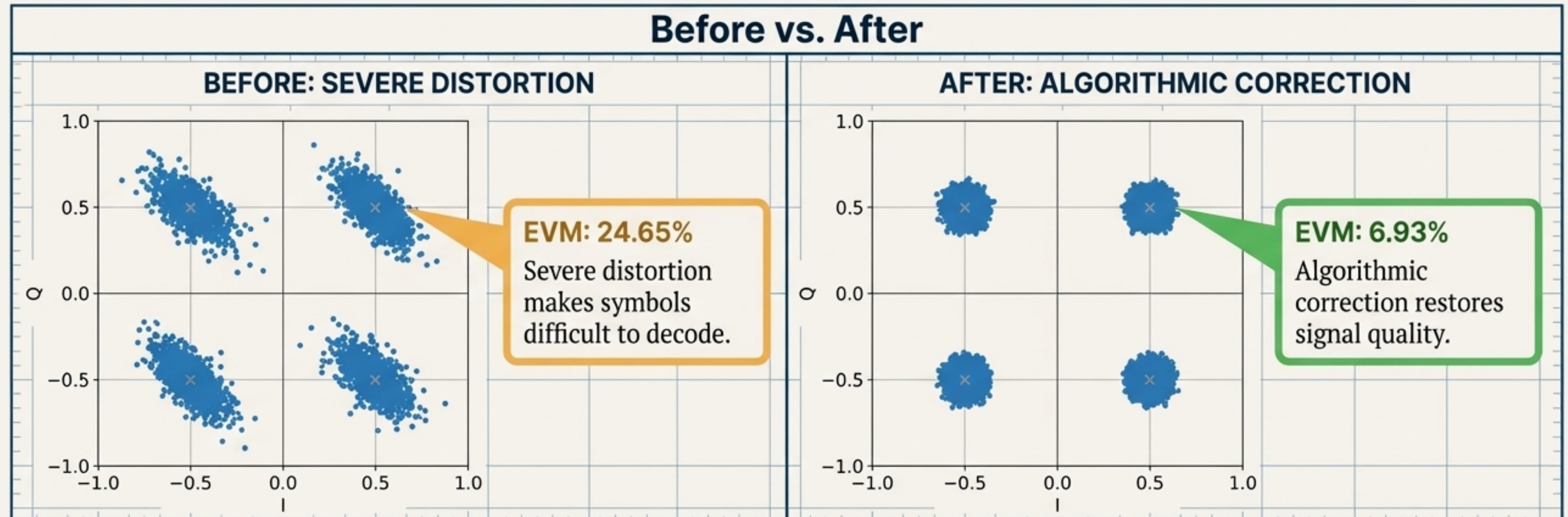
Real-world hardware is not ideal. Mismatches in gain and phase between the I and Q channels distort the signal, leading to constellation skew, image interference, and degraded performance.

The Impact

This distortion is quantified by Error Vector Magnitude (EVM). A heavy imbalance (+20% gain, +15° phase error) can increase EVM from a baseline of 6.9% to a critical 24.6%—a 17.7% absolute degradation.

The Solution

Digital compensation techniques, such as adaptive equalizers (e.g., MA-DFE, PA-DFE), can estimate the imbalance and apply a correction matrix to restore signal integrity.



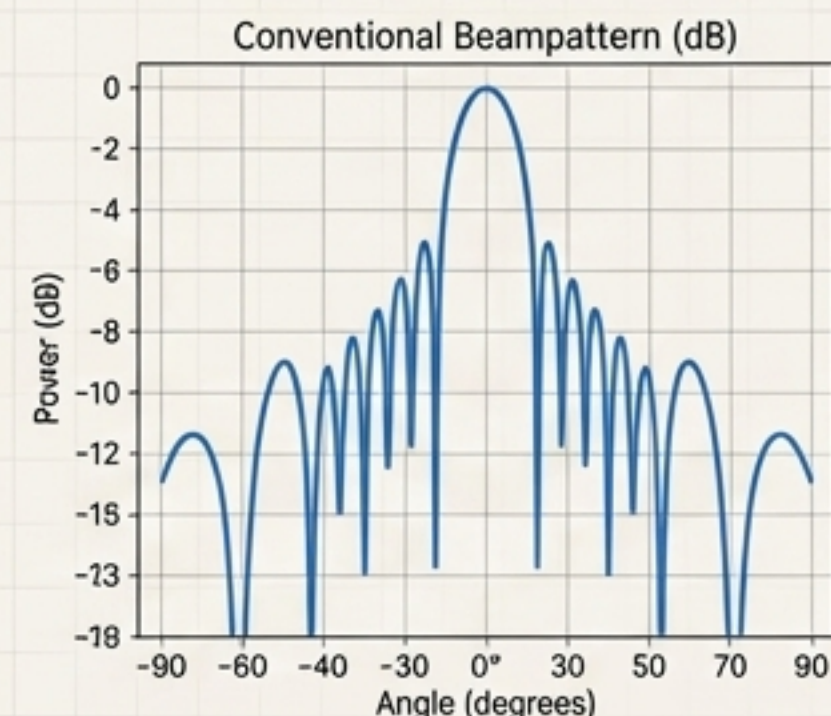
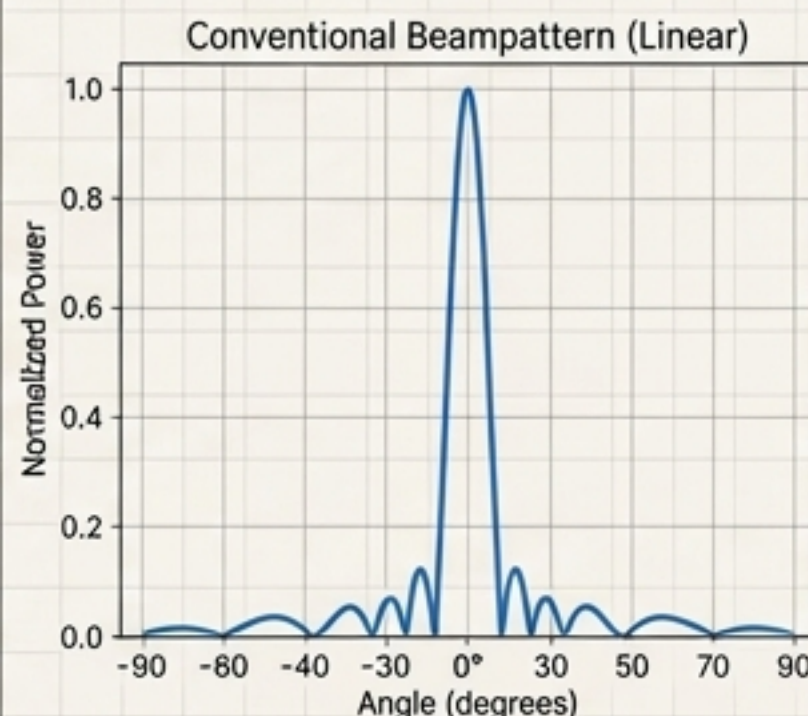
Perceiving Direction: The Array Manifold and Steering Vectors

Core Concept

An antenna array determines a signal's direction of arrival (DoA) by measuring the tiny phase shift of the signal as it arrives at each element. The **Array Manifold**, or steering vector $a(\theta)$, is the mathematical model of this phase response for every possible angle.

How it Works

The steering vector $a(\theta) = [1, e^{j\psi}, e^{j2\psi}, \dots]$ is a series of complex exponentials representing the phase progression. To 'listen' to a specific direction, the system applies weights that are the conjugate of the steering vector, aligning the phases for constructive interference from that direction.

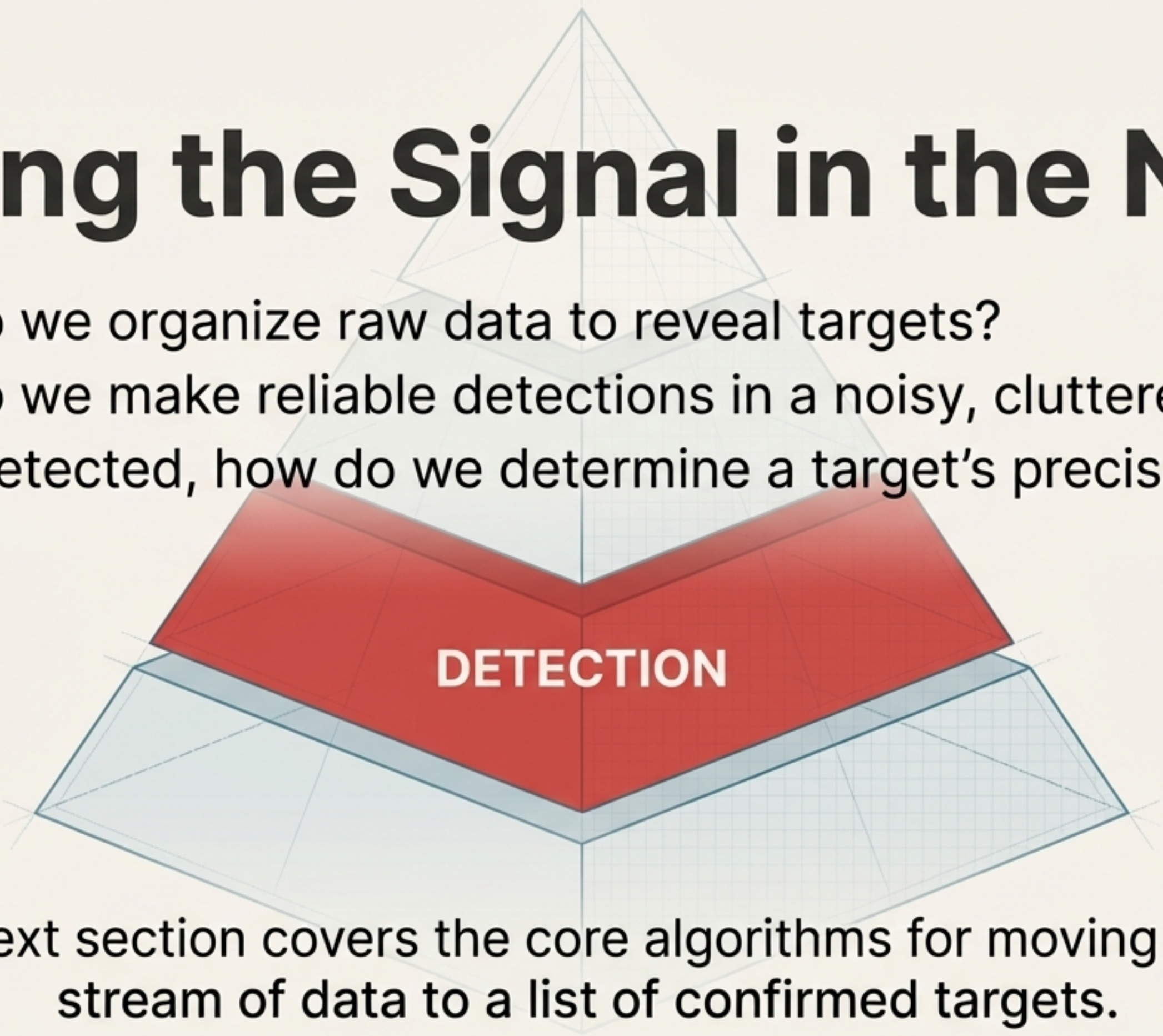


For a signal at 0° , all elements receive in-phase. For 30° , a clear, progressive phase shift is observed. The system learns to map these phase signatures to specific directions.



Finding the Signal in the Noise

- * How do we organize raw data to reveal targets?
- * How do we make reliable detections in a noisy, cluttered world?
- * Once detected, how do we determine a target's precise location?



DETECTION

The next section covers the core algorithms for moving from a stream of data to a list of confirmed targets.

The Detection Landscape: Crafting the Range-Doppler Map

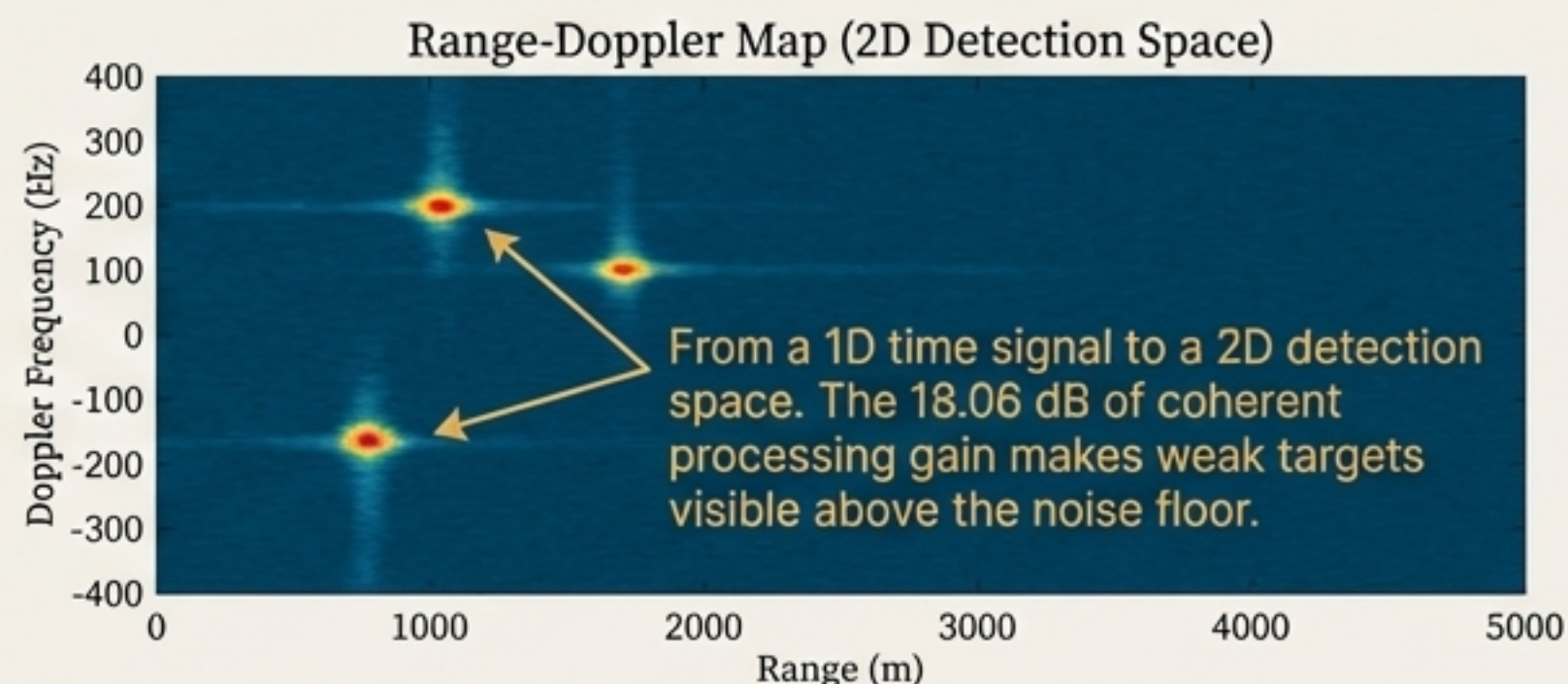
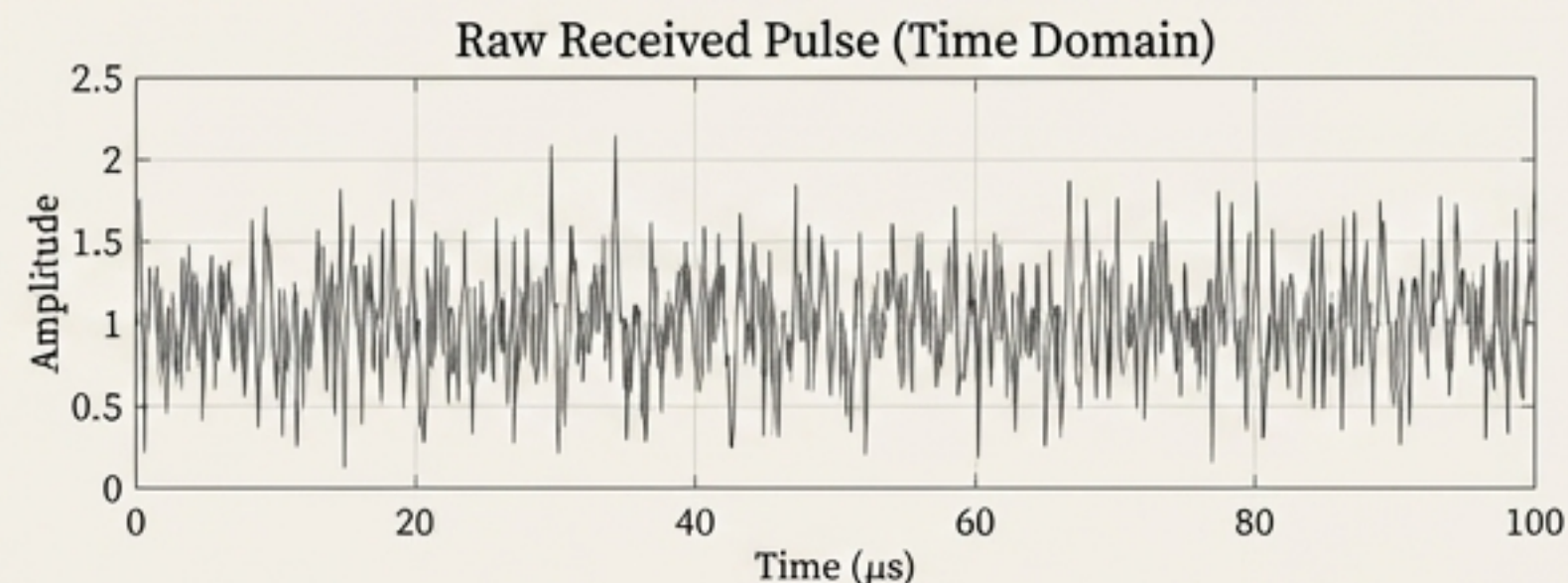
The Process

A Range-Doppler Map (RDM) is a 2D visualization that simultaneously resolves targets in both distance and velocity. It is constructed in two steps:

- 1. Range Compression:** A matched filter is applied to each received pulse to maximize the signal-to-noise ratio and achieve fine range resolution ($\Delta r = c / 2B$).
- 2. Doppler Processing:** An FFT is performed across the coherent train of pulses for each range bin. This integrates the signal energy and resolves the target's velocity ($\Delta f = 1 / LT_p$).

The Result

Targets appear as bright peaks in the RDM, where the X-axis represents range and the Y-axis represents Doppler frequency (velocity). This transforms the detection problem into a peak-finding exercise.



The Shifting Noise Floor: Adaptive Detection with CFAR

The Challenge

A fixed detection threshold is ineffective. In reality, the noise and clutter floor varies dramatically across the Range-Doppler Map. A threshold set too low causes a flood of false alarms; set too high, it misses real targets.

The Method: Cell-Averaging CFAR

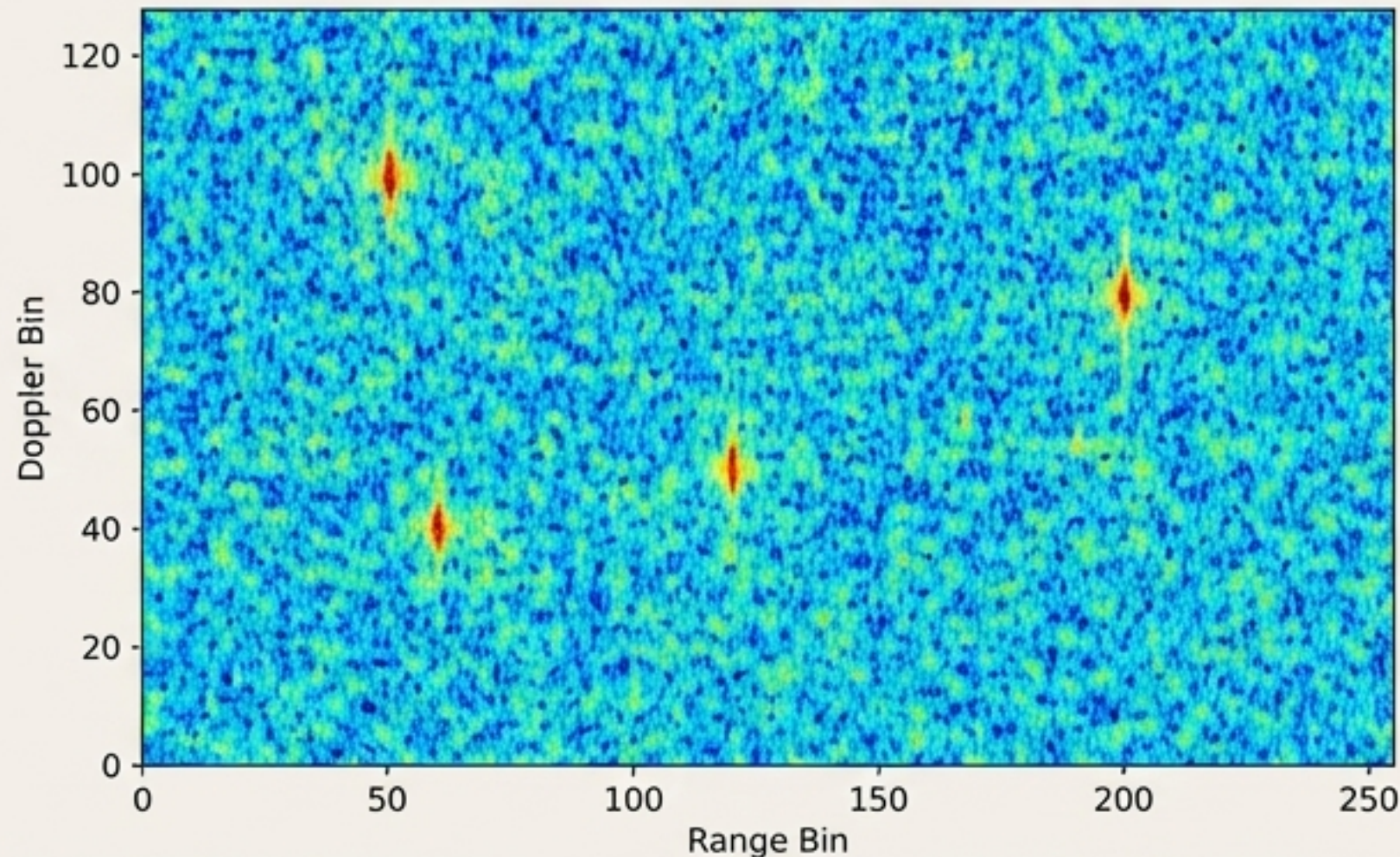
The algorithm creates a dynamic threshold for each cell under test. It works by:

1. Defining a sliding window with reference cells and guard cells.
2. Estimating local noise power by averaging the reference cells.
3. Setting the threshold at a calculated offset (α) above this local estimate.

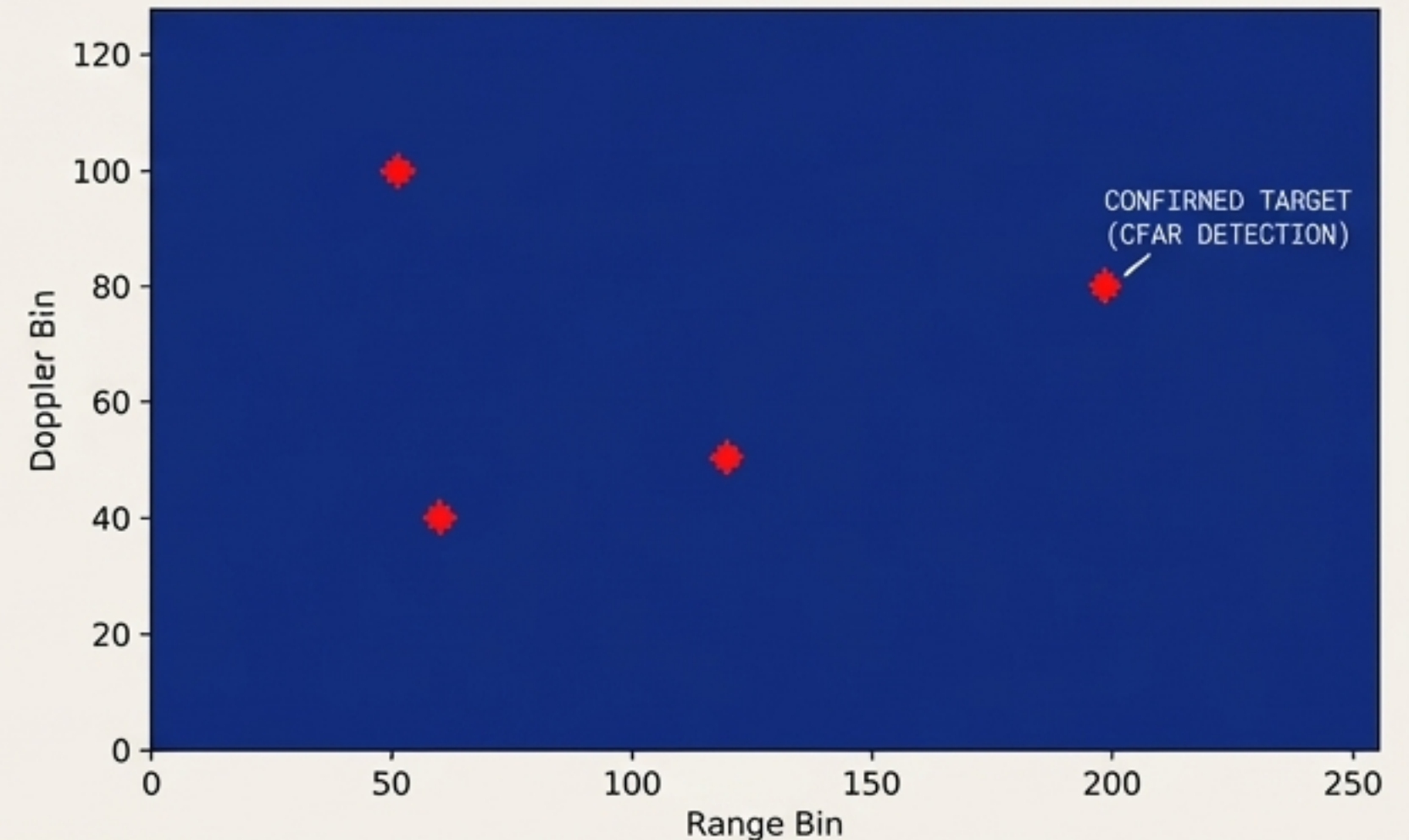
The Result

CFAR maintains a **constant, predictable probability of false alarm (PFA)** across the entire map, reliably detecting targets against a fluctuating background.

BEFORE: NOISY DETECTION MAP



AFTER: CFAR DETECTIONS



Pinpointing the Target: Localization with TDoA

The Challenge

A single radar provides range, but not a unique 2D or 3D position.

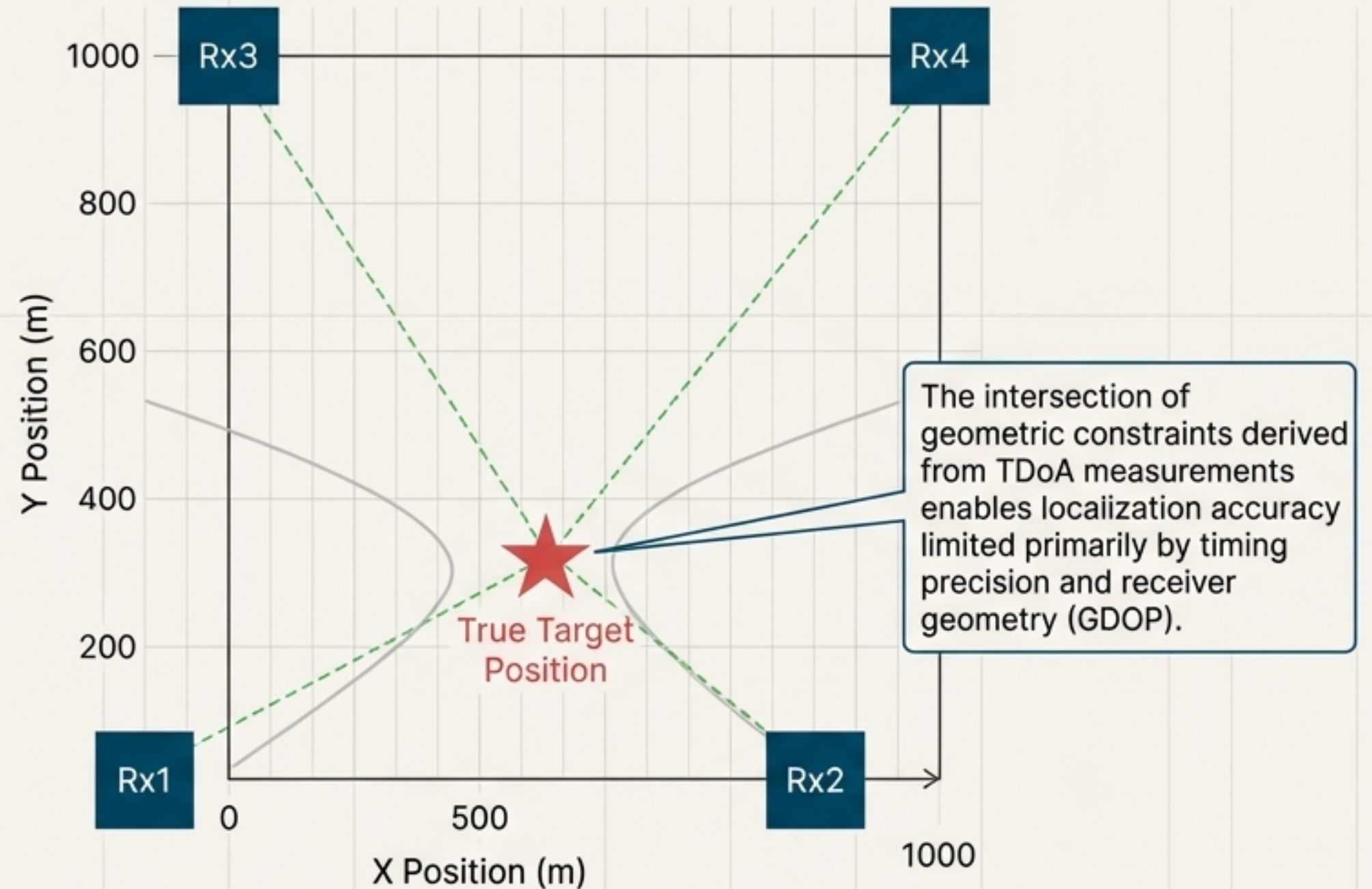
The Method

TDoA uses multiple, spatially separated receivers. The difference in a signal's arrival time at any two receivers confines the target's location to a hyperbola. By using three or more receivers, the intersection of multiple hyperbolas provides a precise position estimate.

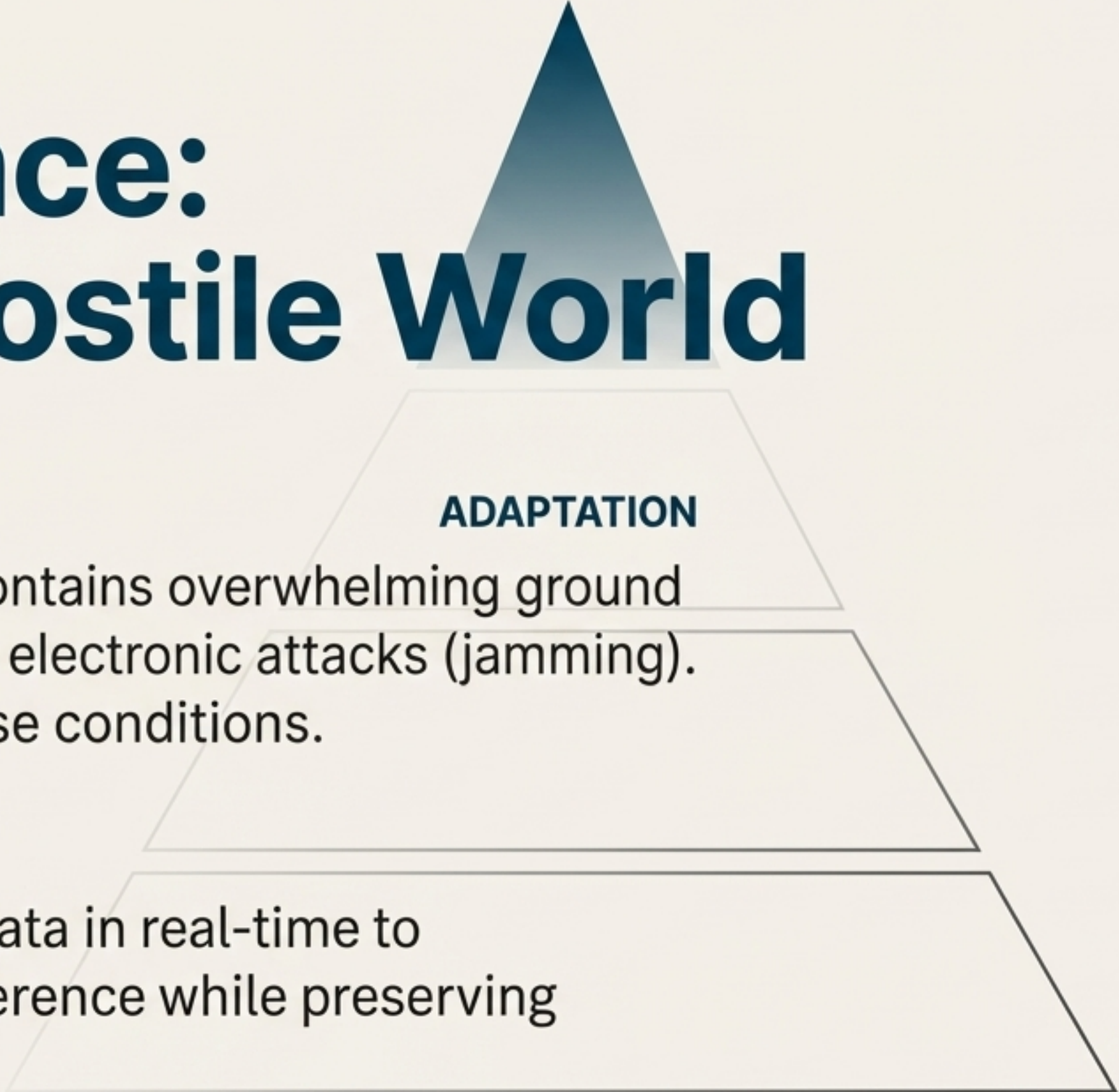
The Algorithm

Chan's Algorithm provides an efficient, closed-form solution to solve the system of hyperbolic equations, converting measured time differences into a coordinate.

Receiver Network and Target Geometry



Peak Performance: Adapting to a Hostile World



The New Reality

The operating environment is not static. It contains overwhelming ground clutter, evolving interference, and deliberate electronic attacks (jamming). Fixed, pre-computed processing fails in these conditions.

The Path Forward

Advanced algorithms must learn from the data in real-time to dynamically create filters that cancel interference while preserving the target signal.

Cutting Through the Clutter: Adaptive Cancellation

The Challenge

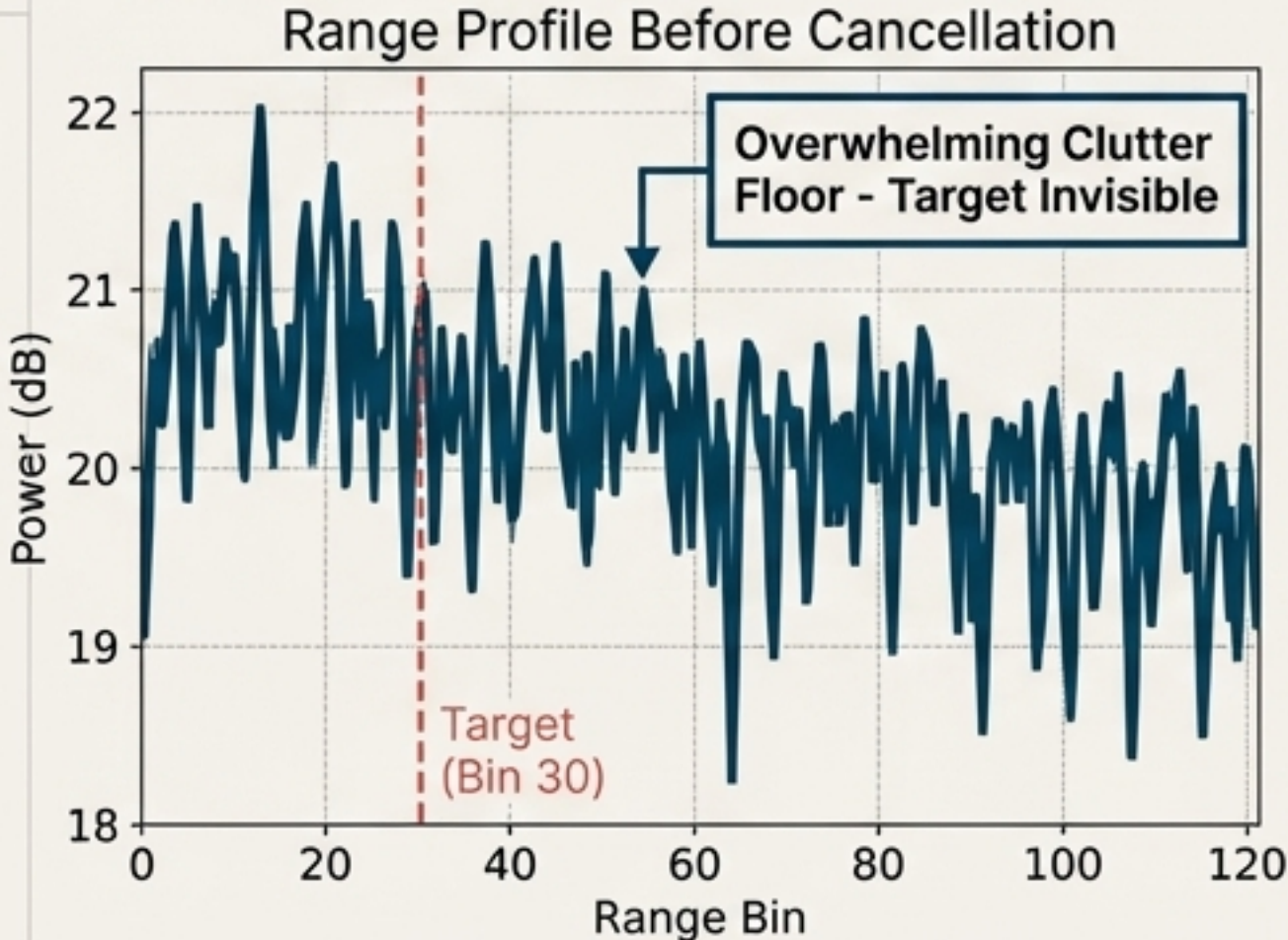
Clutter—unwanted echoes from terrain, buildings, or sea surfaces—can have power levels over 1000x greater than a target's echo. This masks targets completely, rendering the system ineffective. The signal-to-clutter ratio can be as low as -10 dB.

The Method

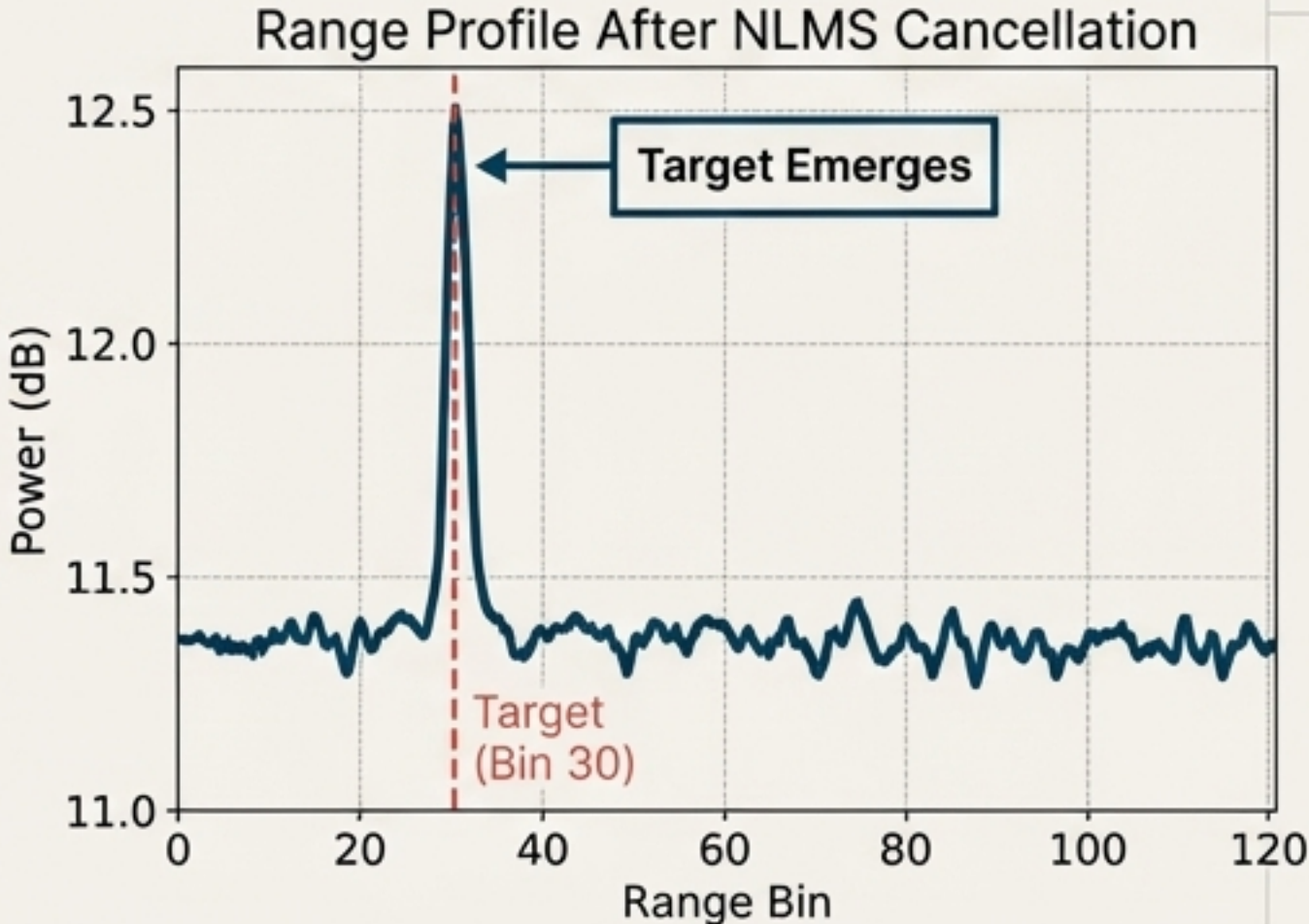
Adaptive filters like the Normalized Least Mean Squares (NLMS) algorithm learn the statistical structure of the clutter from the received data. Because clutter is predictable, the algorithm can create a model of it and subtract it from the signal.

The Result

The target emerges from the noise floor.



8.31 dB
Clutter Rejection Ratio (CRR)
achieved, making
the target detectable.



Defeating the Jammer: Null-Steering with Adaptive Beamforming

The Challenge

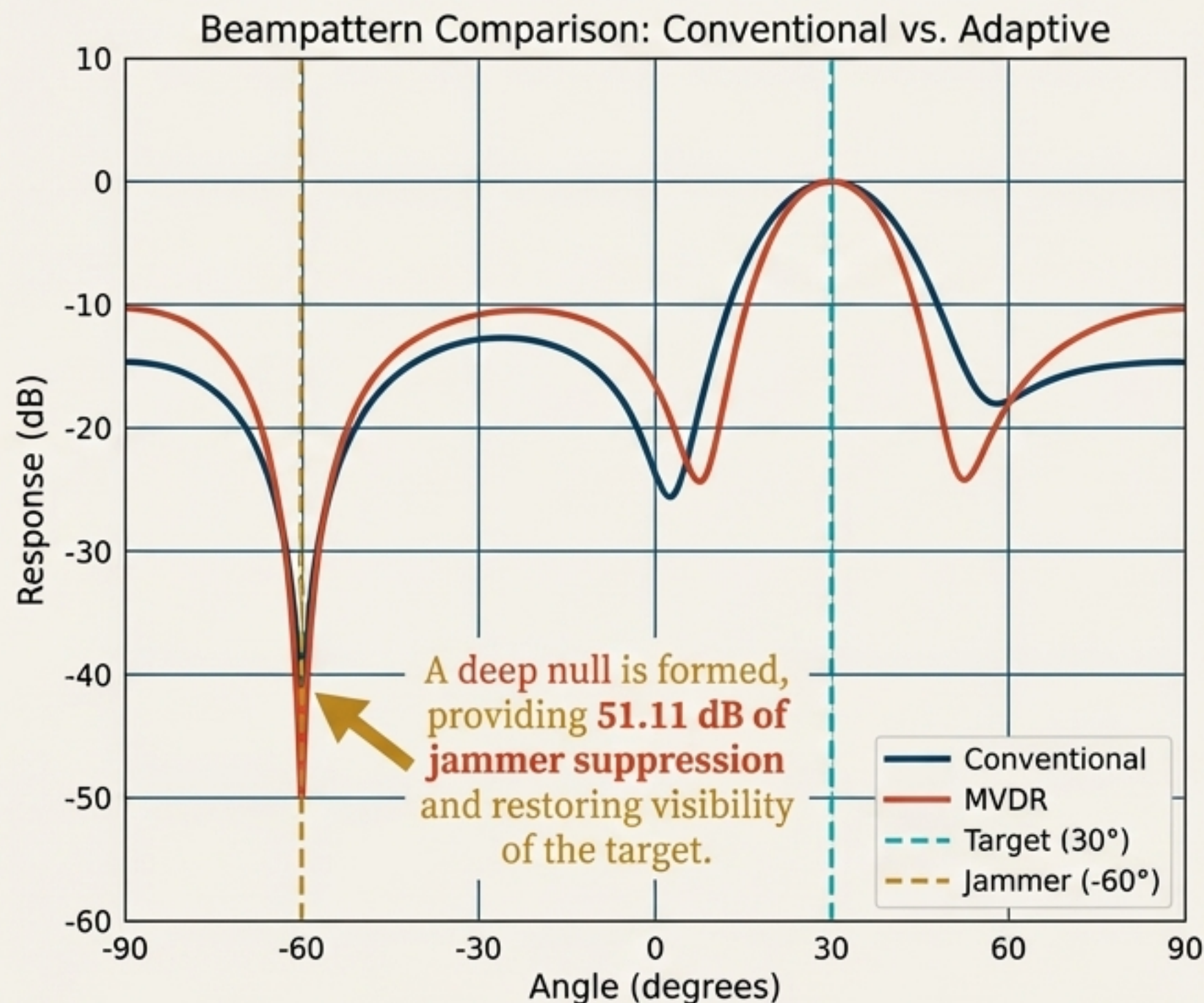
A strong jammer can overwhelm a radar's receiver, blinding it in a specific direction and masking any potential targets. A conventional beamformer is helpless against this.

The Method: MVDR

The Minimum Variance Distortionless Response (MVDR) beamformer is an optimal adaptive algorithm. Its goal is to minimize the total power at the array's output, under the single constraint that it must maintain unit gain in the target's direction. This has the natural effect of seeking out and placing infinitely deep nulls on the most powerful signals—the jammers.

The Result

The array forms a beampattern that preserves the target while creating a surgical null precisely on the jammer, restoring the system's ability to see.



The Ultimate Defense: Space-Time Adaptive Processing (STAP)

The Challenge

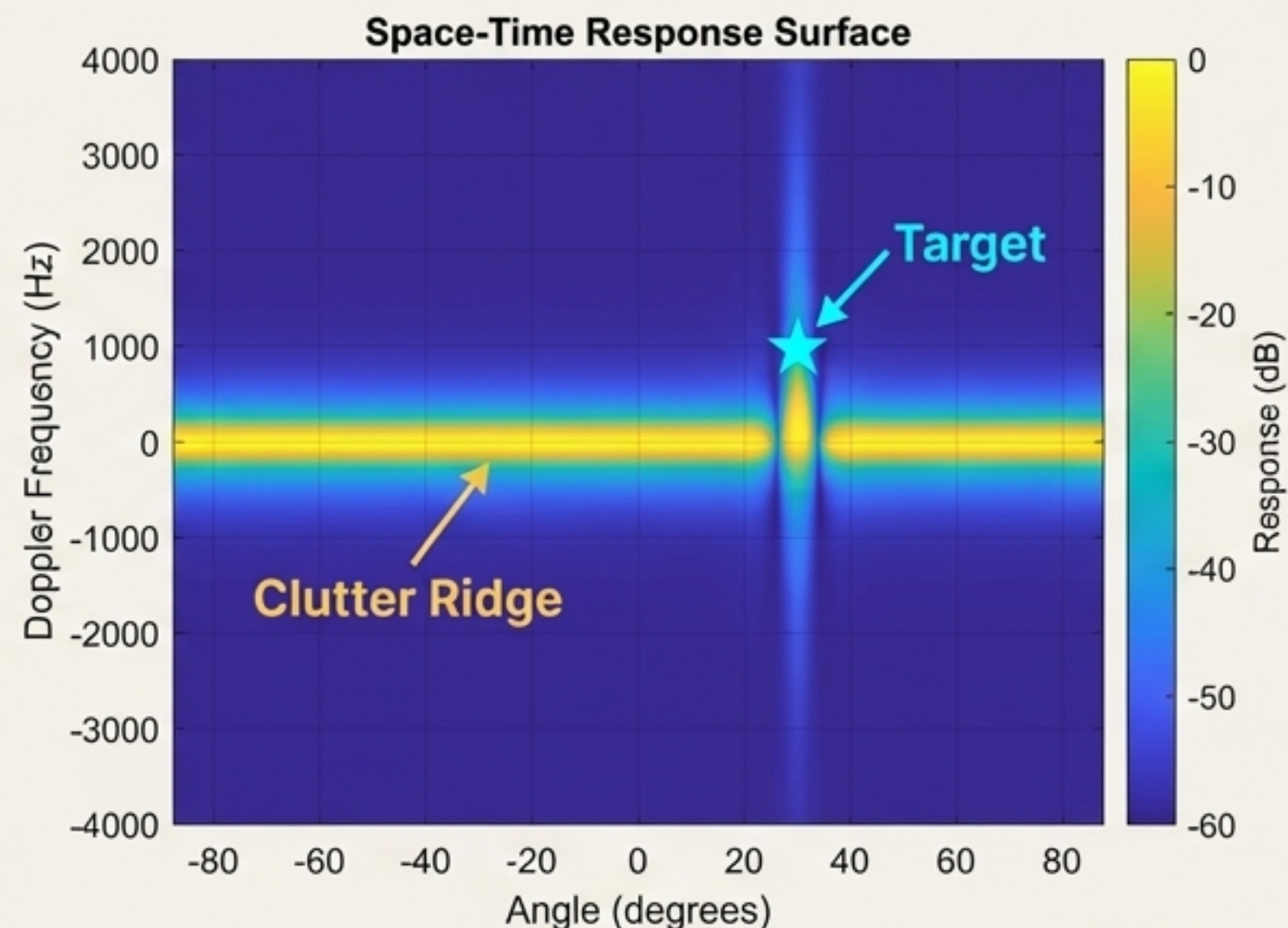
Real-world clutter is not just stationary (zero Doppler) or from one direction. Moving platforms induce a relationship between clutter angle and Doppler, creating a "clutter ridge" in the Range-Doppler map that can obscure targets.

The Method

STAP jointly processes data across both the spatial dimension (M antenna elements) and the temporal dimension (L pulses). It forms a space-time covariance matrix and designs an optimal $M \times L$ dimensional filter to place a null along the entire clutter ridge while preserving gain on a specific angle-Doppler cell.

The Result

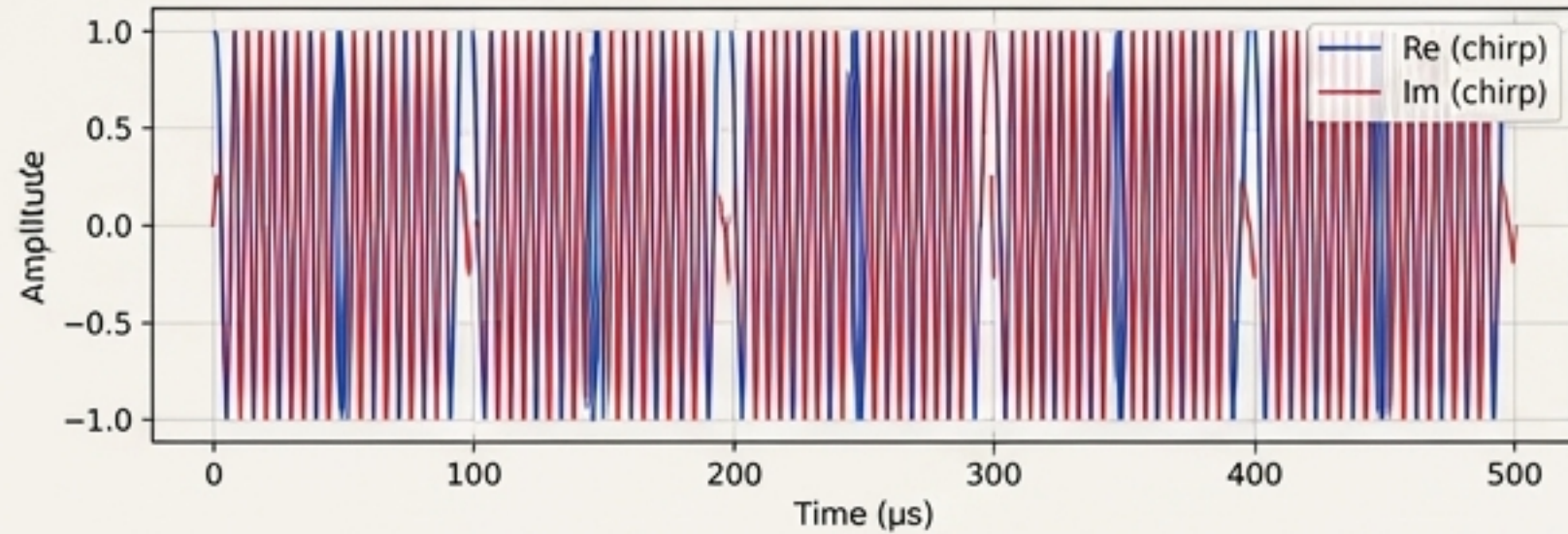
A 2D filter that is precisely tuned to the specific clutter environment, offering superior rejection of distributed clutter.



STAP creates a **2D filter** that nulls the entire clutter ridge (yellow line) while maintaining high gain on the target (cyan star), achieving **SINR improvements of 40-70 dB** in practice.

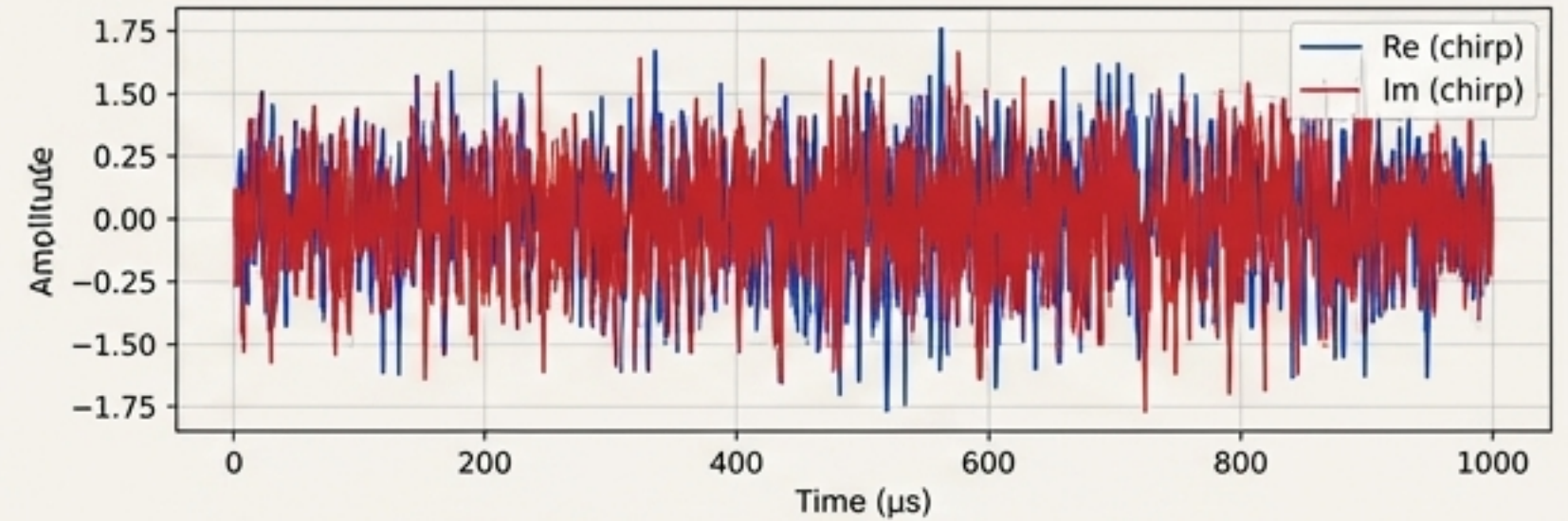
The Apex: A Complete End-to-End Processing Chain

Step 1: Transmitted Signal



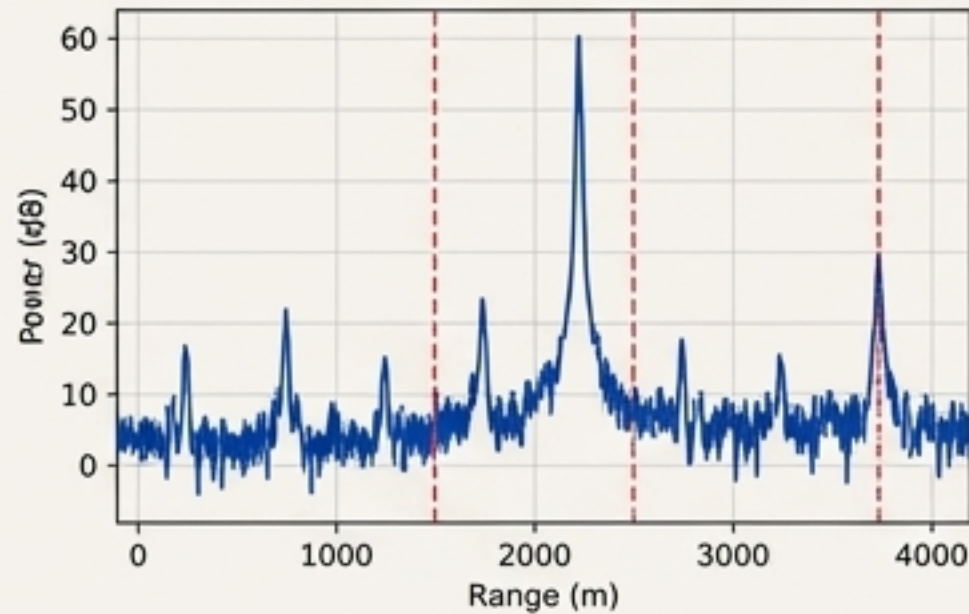
The signal we send.

Step 2: Received Raw Pulse



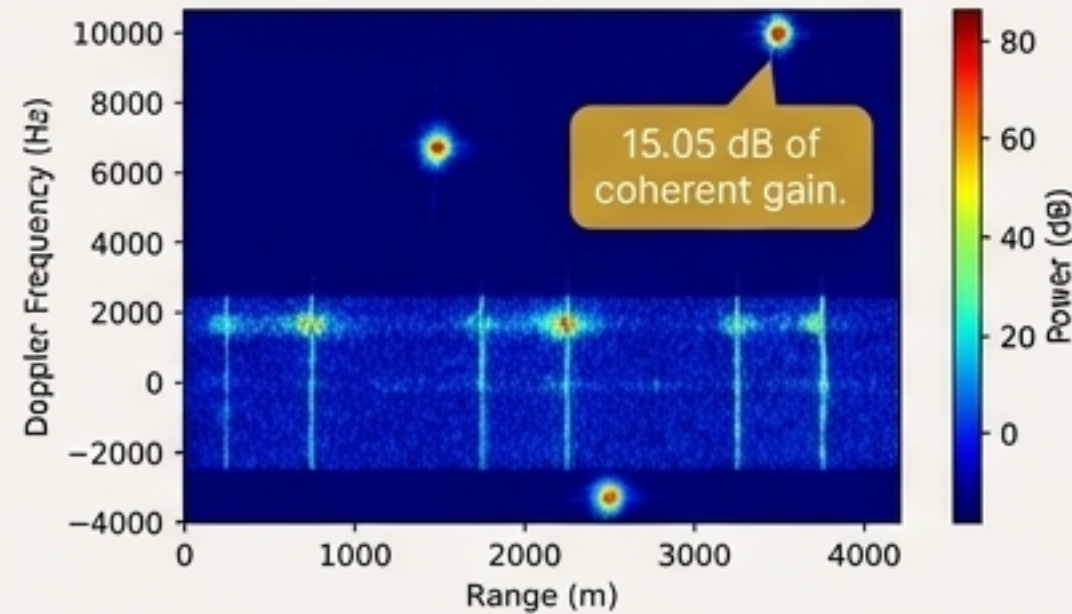
The noisy, complex signal we get back.

Step 3: Range Compression



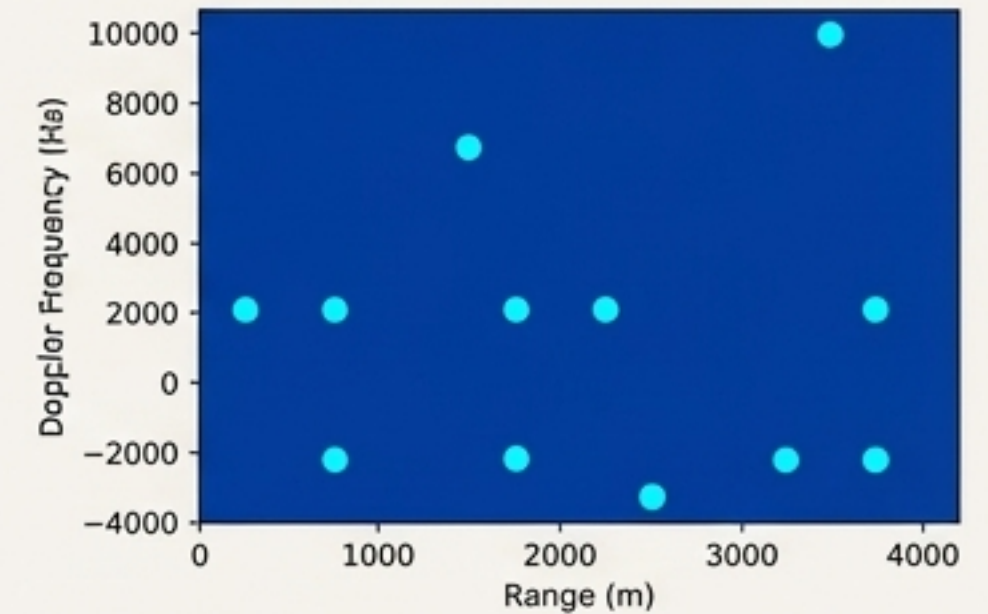
Clear peaks emerge at target ranges.

Step 4: Range-Doppler Map



The fully processed detection space, revealing targets in range and velocity with 15.05 dB of coherent gain.

Step 5: Final Detections



The final output: a clean list of detected targets, ready for tracking.

A Framework for Capability: Algorithm Selection and Trade-offs



Final Takeaway

The key to modern radar performance lies in integration. Each algorithm solves a specific problem, from signal representation to interference suppression. Success comes not from a single “best” algorithm, but from architecting an adaptive chain that learns from its environment to overcome real-world challenges.

Practical Guidance: Algorithm Trade-offs

Algorithm	Time Complexity	Latency	Real-Time Feasible?	Best For
Range-Doppler Map	$O(NL \log(NL))$	$\sim 10 \text{ ms}$	Yes	Baseline detection
LMS Beamforming	$O(M)$ per sample	$\sim 100 \mu\text{s}$	Yes	Low-cost adaptation
MVDR Beamforming	$O(M^3)$ per update	$\sim 10 \text{ ms}$	Yes (if $M \leq 16$)	Optimal jammer nulling
STAP (Full)	$O((ML)^3)$	$\sim 1 \text{ s}$	No	Theoretical best
STAP (Factored)	$O(M^3 + L^3)$	$\sim 100 \text{ ms}$	Yes (with GPU)	High-performance clutter rejection